UBER EDA INSIGHTS REPORT

Supply, demand, trip outcomes and operational observations

Project: Uber EDA Project

Project type: Exploratory Data Analysis (EDA)

Contribution: Individual

Name: Sanketh S

Business focus

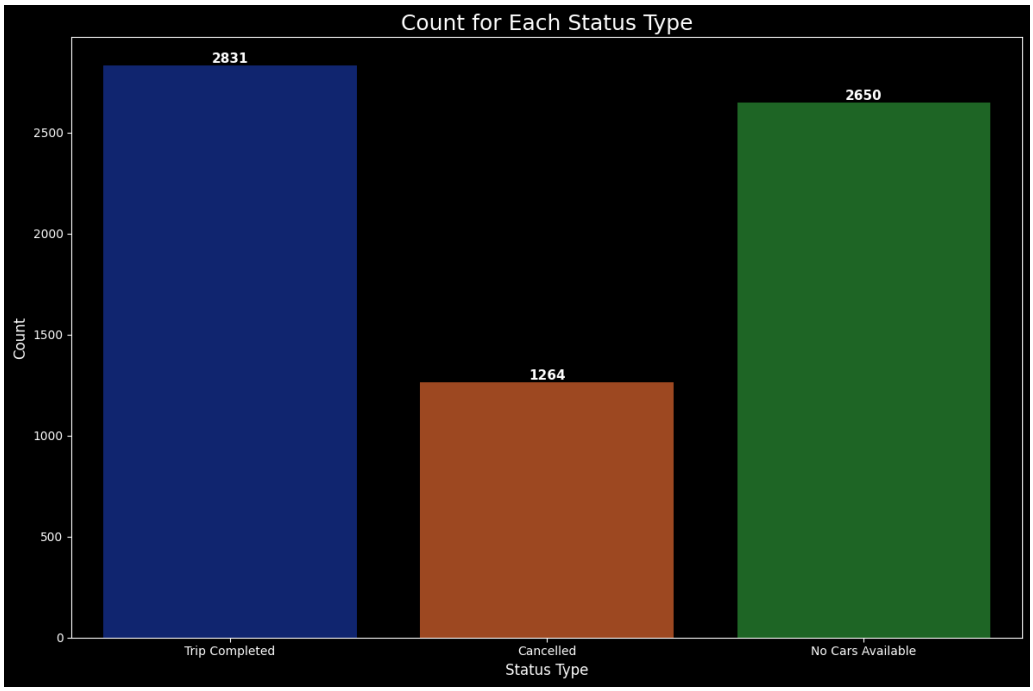
The analysis examines customer requests, pickup locations and trip outcomes to identify where demand is concentrated and where service availability or cancellations may be limiting successful trips.

> Dataset snapshot: 6,745 records and 6 columns. The notebook reports no duplicate records. Driver ID and Drop timestamp contain missing values, which were handled by replacing missing entries with “Not Available”.

Business objective

Identify demand patterns and unsuccessful trip outcomes so that vehicle supply, driver allocation and operational planning can be improved at the locations and periods where customers are most affected.

1. Overall Trip Status



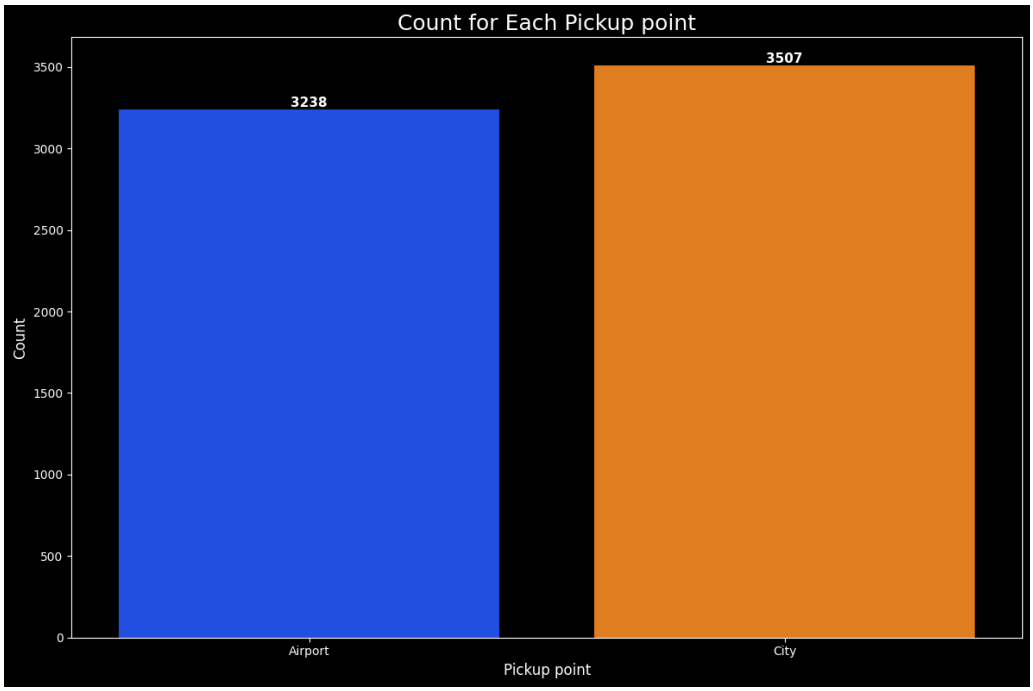
Why this chart? A count plot provides a quick view of how requests are distributed across the three main outcomes.

Insight: The notebook chart shows 2,831 completed trips, 1,264 cancelled trips and 2,650 requests with no cars available. This means a substantial share of requests did not result in a completed trip.

> Business impact: The unsuccessful outcomes represent opportunities for improvement. Reducing cancellations and vehicle-unavailability cases can increase completed trips and improve customer experience.

Action: Track unsuccessful requests separately and investigate the operational reasons behind each category.

2. Pickup Point Demand



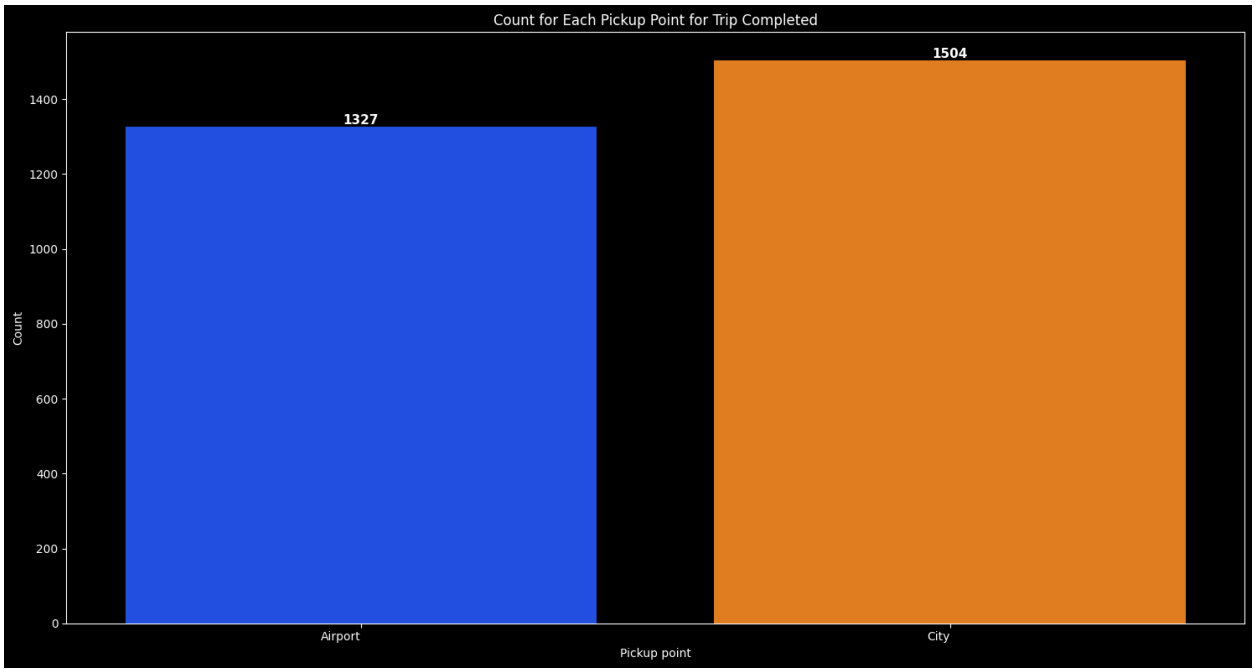
Why this chart? Comparing pickup-point counts shows where requests are concentrated.

Insight: City accounts for 3,507 requests while Airport accounts for 3,238. Both locations generate high demand, with City slightly higher in the overall request volume.

> Business impact: Supply planning should consider both pickup points instead of treating demand as evenly distributed.

Action: Use pickup-point demand as a baseline for vehicle positioning and driver allocation.

3. Successful Trip Completion



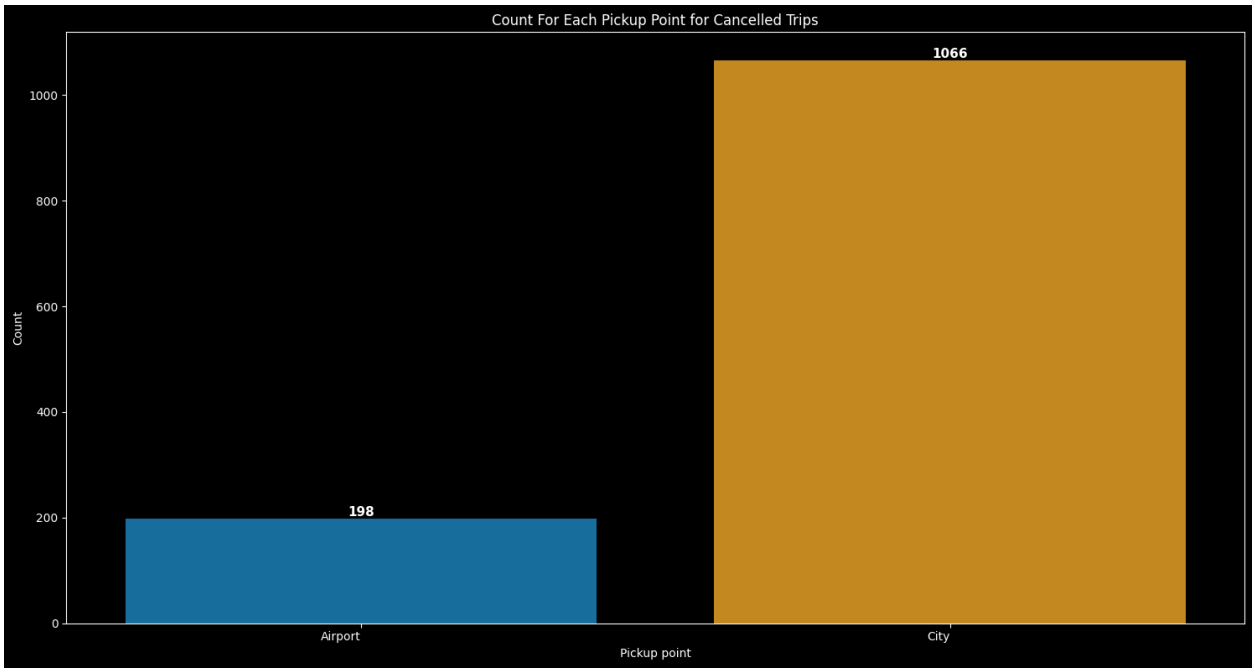
Why this chart? Completed-trip counts show which pickup locations convert requests into successful rides.

Insight: The chart shows 1,327 completed trips from Airport and 1,504 from City. Although City has slightly higher overall demand, it also records more completed trips.

> Business impact: The difference between total requests and completed trips is important because it highlights where additional supply or better operations may increase successful service.

Action: Compare completion counts with cancellations and no-car cases at each pickup point.

4. Cancellation Pattern



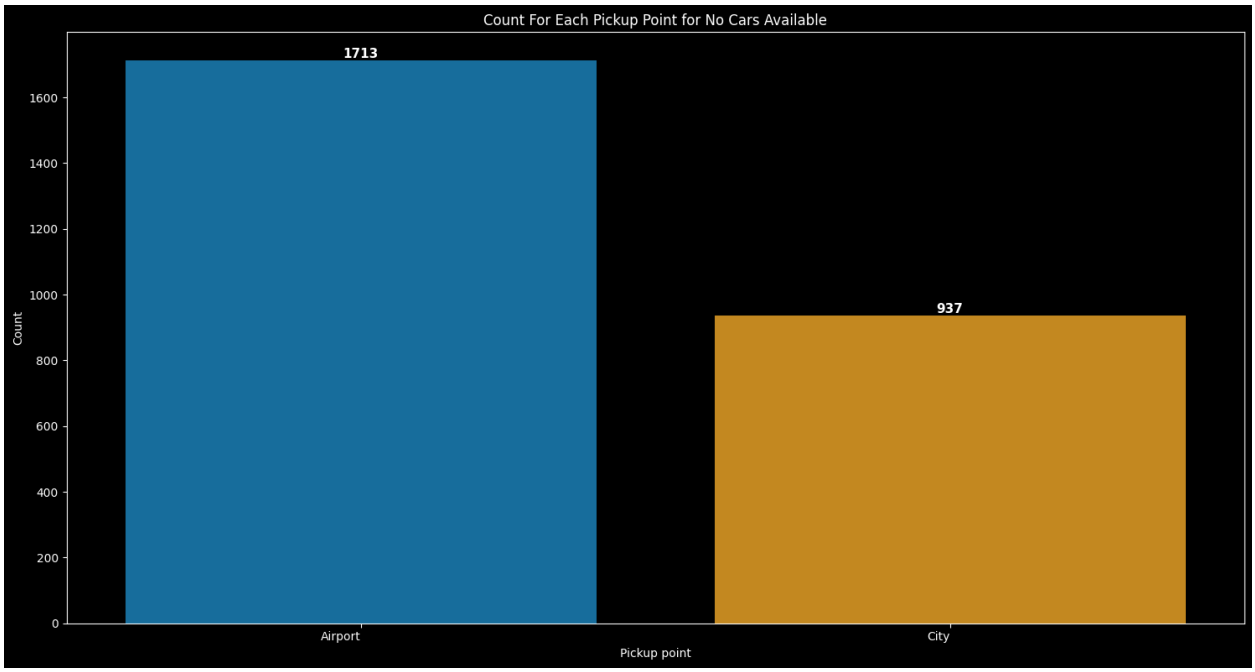
Why this chart? A location-wise cancellation comparison helps identify where customer requests are being lost after booking.

Insight: The chart shows 1,066 cancelled trips at Airport and 198 in City. Airport therefore has a much larger cancellation count in this dataset.

> Business impact: A high cancellation count can reduce successful rides, revenue opportunities and customer satisfaction.

Action: Examine driver availability, waiting time and assignment efficiency around Airport requests.

5. Vehicle Availability Gap



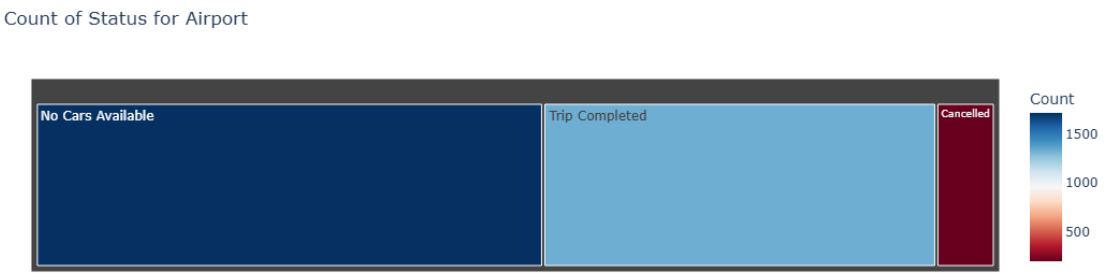
Why this chart? This comparison directly highlights where customers are unable to obtain a vehicle.

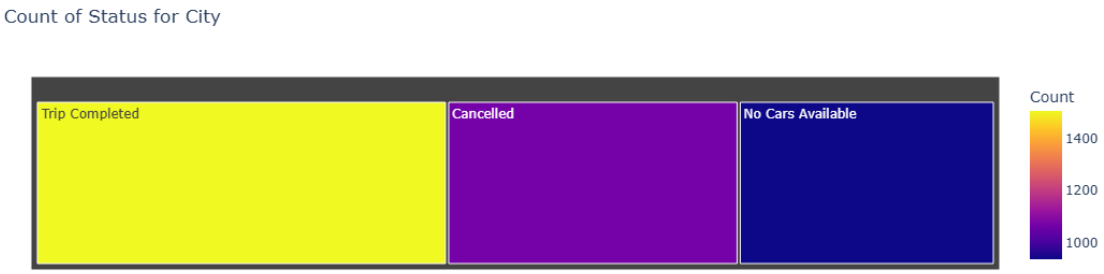
Insight: The notebook chart shows 846 'No Cars Available' cases at Airport and 1,804 in City. City therefore has the larger vehicle-unavailability problem.

> Business impact: These cases represent requests that fail before a trip can begin. Improving availability could directly increase the number of completed trips.

Action: Increase or reposition supply during periods when City demand is high and vehicles are unavailable.

6. Location-Specific Status View



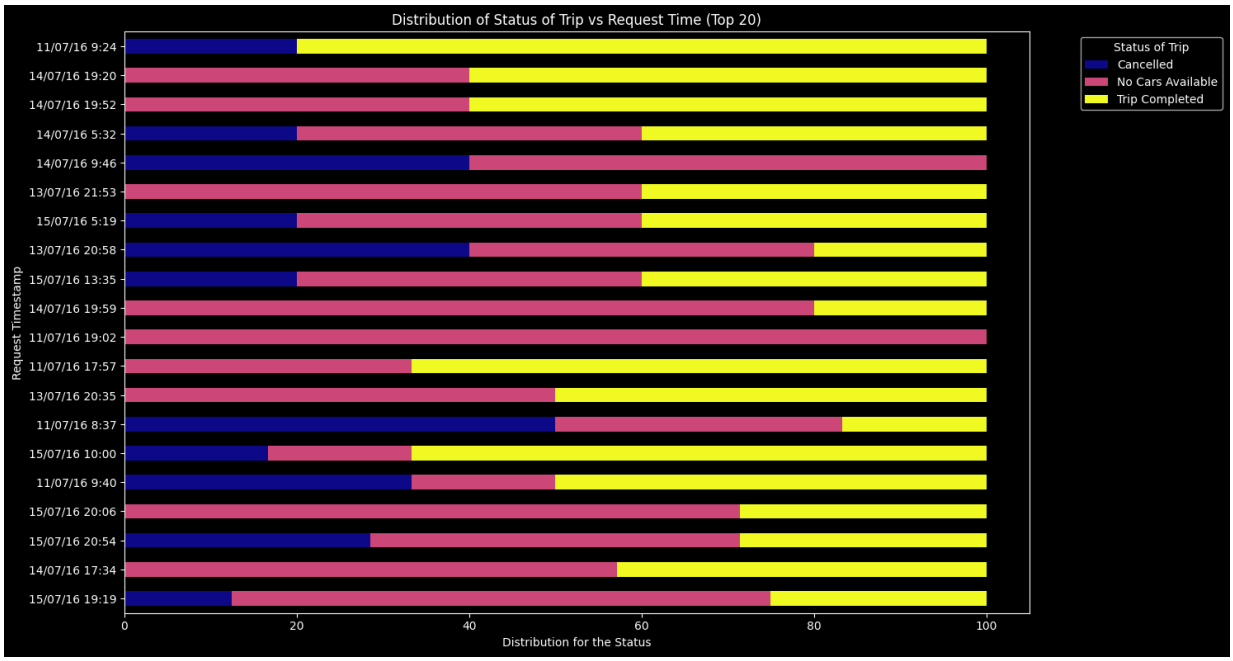


Why these charts? The location-specific views separate Airport and City outcomes, making it easier to see whether the operational problem is the same at both pickup points.

Insight: The Airport view emphasizes the balance among completed, cancelled and no-car outcomes, while the City view shows a stronger presence of completed trips alongside a notable no-car segment.

> Business impact: A location-specific strategy is more useful than applying one supply rule to every pickup point.

7. Time-Based Demand View



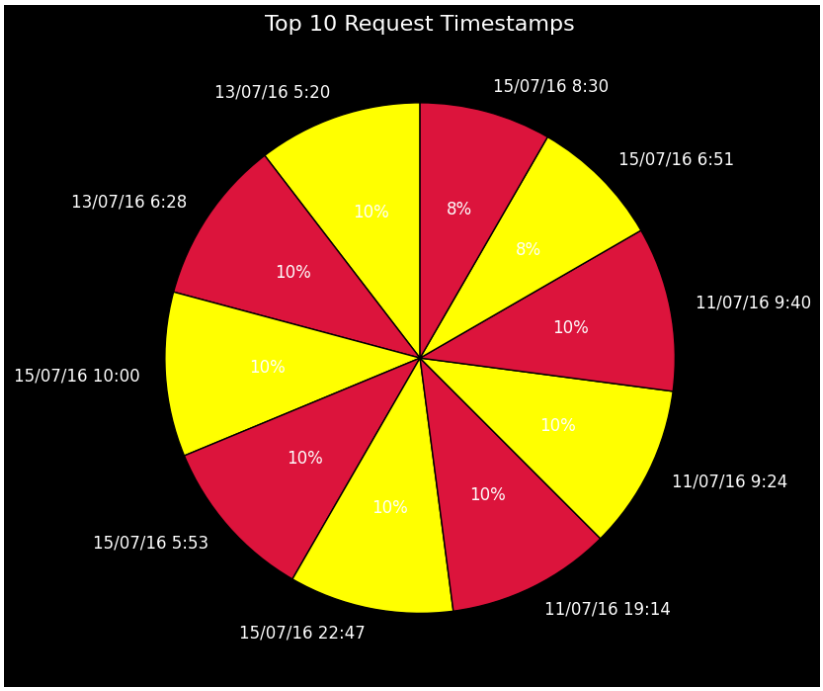
Why this chart? The stacked time-based chart shows how the distribution of pickup points changes across request timestamps.

Insight: The chart demonstrates that request activity varies by time, and the mix of Airport and City requests is not identical at every timestamp.

> Business impact: Time-based demand patterns can support shift planning, vehicle positioning and targeted driver incentives.

Action: Combine these time patterns with cancellation and no-car outcomes to identify the periods where extra supply would have the greatest potential benefit.

8. Peak Request Timestamps

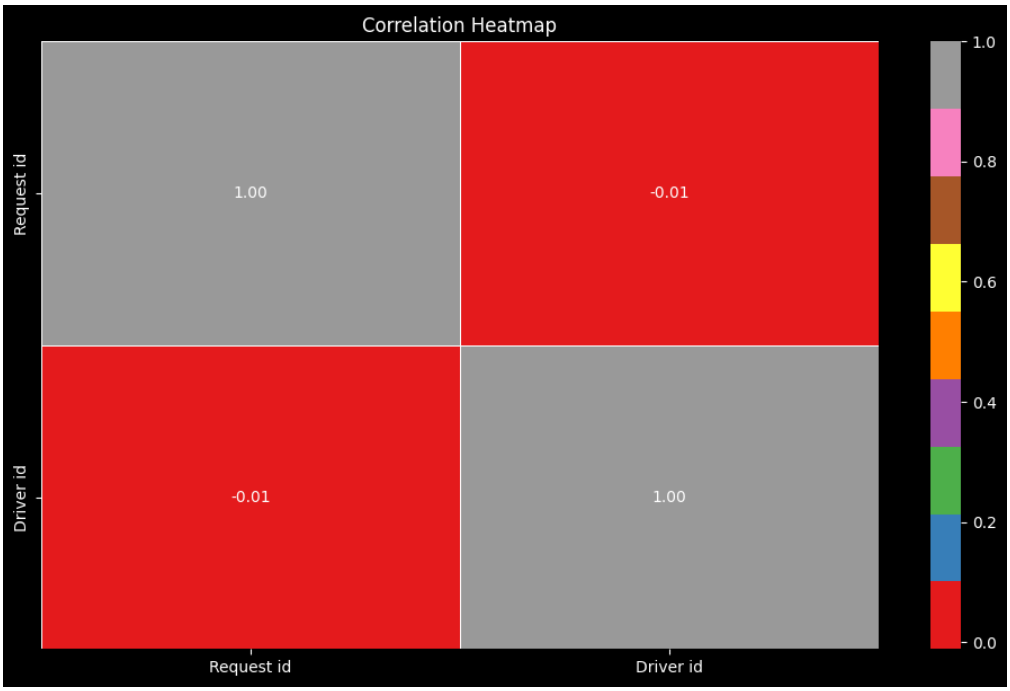


Why this chart? The pie chart highlights the top request timestamps identified in the notebook.

Insight: Requests are concentrated among a limited set of timestamps, showing that demand is not evenly spread across all observed request times.

> Business impact: Concentrated demand creates opportunities for proactive supply planning rather than relying only on reactive driver allocation.

9. Numerical and Relationship Analysis



Interpretation: The notebook's correlation heatmap compares the numerical columns used in the analysis. The relationship between Request ID and Driver ID is not a business relationship in the usual sense because both are identifiers.

> Important caution: ID columns should not be treated like continuous business measures. Their numerical values are labels, so correlation or pair-plot patterns involving IDs should not be used as evidence of operational causation.

Usefulness: The numerical charts are best treated as supporting EDA visuals, while status, pickup-point and time-based charts provide the stronger business insights.

10. Recommended Business Actions

• Balance supply with demand: Use pickup-point and time-based request patterns to position vehicles where demand is expected.

• Reduce vehicle-unavailability cases: Give priority to periods and locations with large 'No Cars Available' counts.

• Investigate cancellations: Airport cancellations are especially prominent in the dataset and should be examined for operational causes.

• Plan by location: City and Airport show different unsuccessful-trip patterns, so they should not necessarily receive identical supply strategies.

• Monitor peak periods: Use the request-time analysis to prepare drivers before demand peaks instead of responding only after shortages occur.

• Measure improvement: Track completed, cancelled and no-car counts over time to confirm whether interventions are improving successful-trip conversion.

Final Conclusion

The EDA shows that Uber's request outcomes are strongly connected to service availability and location-specific operating conditions. The dataset contains a large number of unsuccessful requests through cancellations and 'No Cars Available' outcomes. City has slightly higher overall request volume and a larger no-car count, while Airport has a much higher cancellation count. These differences suggest that supply planning should be based on both where and when customers request rides.

> Overall, the analysis can help the business focus on the most actionable areas: improving vehicle availability, reducing cancellations, and using time-based demand patterns for proactive driver allocation.

Note: The charts included in this report are taken from the submitted EDA notebook. The written insights are based on the values and visual patterns supported by those notebook outputs.